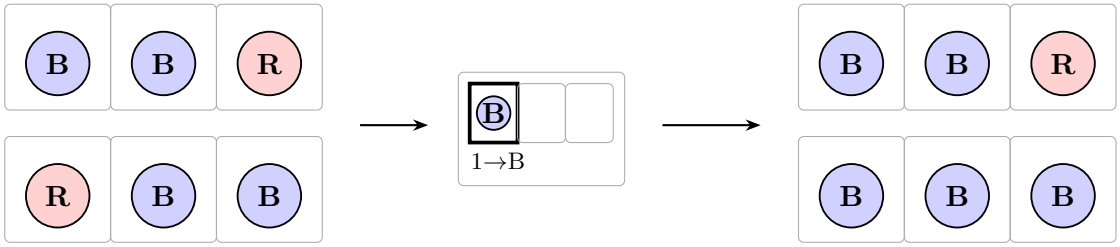
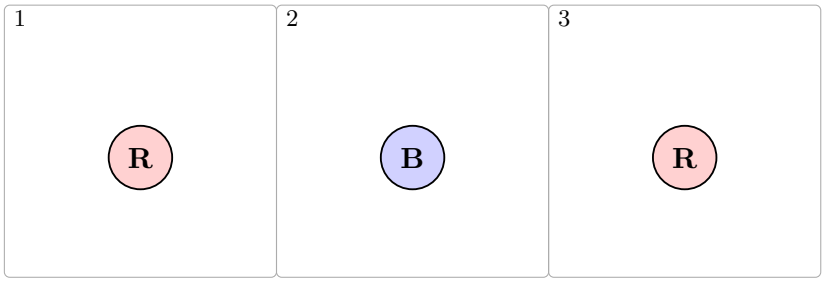
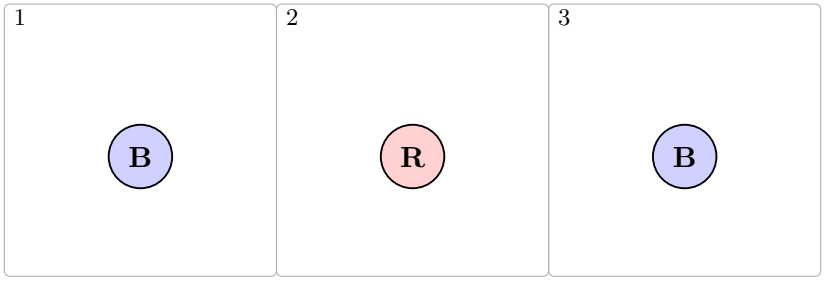
Each instruction names a slot and a color. Set that slot to that color on both boards, even if it is already that color. Slots are 1, 2, 3 from left to right.




**Problem 1:** Find the fewest instructions needed to make these boards match. Build pairs needing fewer instructions. Can any pair need more?



**Problem 2:** Which stories guarantee that any two starting boards will finish matching? Explain why your answer holds for every starting pair.



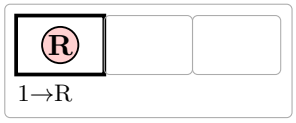


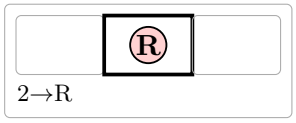


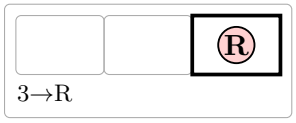


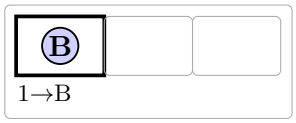

**Problem 3:** Beside each story, mark a slot the first time the story names it. Can two different starting boards match before all three slots are marked? Make one example that does and one that does not.

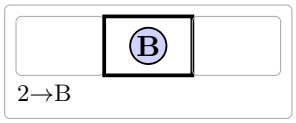
Mix six instruction cards in a cup. Draw, use the instruction on both boards, replace, and mix.

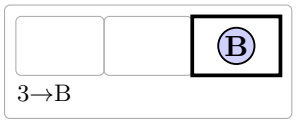


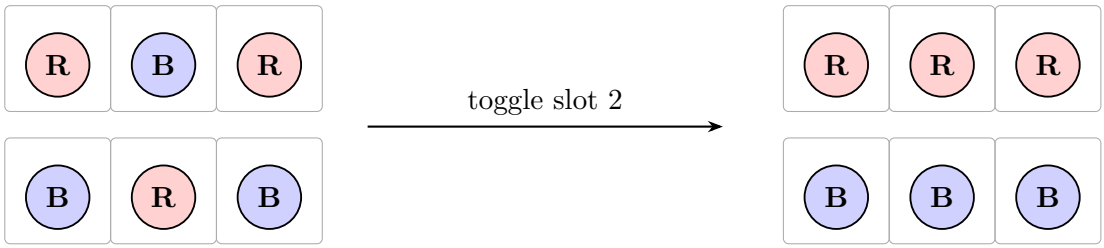



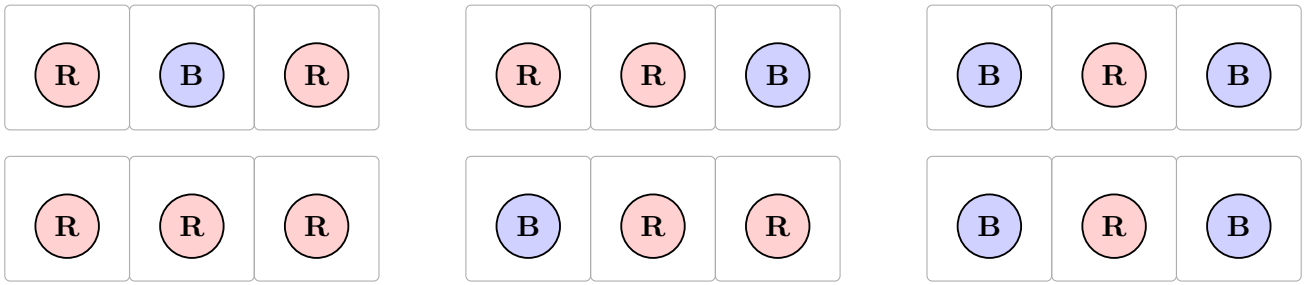

**Problem 4:** A friend says three draws must make every pair match. Make a three-draw story that tests that claim. Could even ten draws fail?

**Problem 5:** Find the shortest instruction story that guarantees a match for every starting pair. Explain why no shorter story can give that guarantee.

A toggle changes red to blue or blue to red in its slot, on both boards. It never sets a chosen color.




**Problem 6:** Use only shared toggles on these starting pairs. Which can be made to match, and why?




**Problem 7:** What changes if a story may use both color-setting instructions and toggles? Invent a story that makes every starting pair match, and a story that fails for at least one pair. Show a pair where it fails.